Jeongin Kim

M.S. student in software testing. LLM-driven generation of test environments, and predictive modeling on industrial process data.

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Education

2025.03–present **M.S. Student, Computer Science and Engineering** Daegu, Korea
Kyungpook National University
Advisor: Prof. Woojin Lee, Software Testing Lab (STLAB). Thesis topic not yet fixed.

2023.03–2024.02 **B.Eng., Computer Information Engineering** GPA 3.50/4.50 | Daegu, Korea
Yeungjin University (Advanced Major Course)

2018.03–2023.02 **Associate Degree in Engineering, Web Database** GPA 3.05/4.50 | Daegu, Korea
Computer Information Division, Yeungjin University

2018.10–2020.06 Republic of Korea Army, mandatory military service (completed)

Research Interests

Software testing, LLM-based test generation, SIL verification environments, predictive modeling on industrial process data

Publications

*corresponding author, own name in bold

Domestic Conference Papers

all in Korean

[1] **Jeongin Kim**, Deokyeop Kim, Hyeryung Suh, and Woojin Lee*. Automatic Generation of Test Environment Behaviors Using LLM for Embedded Software Testing. *Proc. Korea Computer Congress (KCC)*, pp. 582–584, June 2026. Code: github.com/99JIK/KCC2026

[2] **Jeongin Kim** and Woojin Lee*. A Study on the Utilization of Sampling Time-point Data for Performance Enhancement of the Biodegradable Fiber Spinning Process. *Symposium on Ubiquitous Computing and Web Information Technology (UCWIT)*, KIISE Yeongnam Branch, November 2025.

[3] **Jeongin Kim** and Woojin Lee*. The Ability of Generative AI to Understand and Utilize Test Cases. *Proc. Korea Computer Congress (KCC)*, pp. 397–399, July 2025.

Patents

[1] **Jeongin Kim**, Deokyeop Kim, and Woojin Lee. A Method for Selecting Robust Optimal Process Parameter Combinations. Provisional filed, full application in progress (July 2026). Applicant: KNU Industry-Academic Cooperation Foundation. MOTIE/KIAT P0022335.

Experience

2024.12–present

Graduate Researcher Daegu, Korea

Software Testing Lab, Kyungpook National University

- LLM-driven generation of environment object behaviors for SIL testing. A three-stage decomposition-induction-synthesis prompt pipeline more than doubled coverage over zero-shot and few-shot baselines across 11 objects in three domains (elevator, drone, smart factory), consistent on GPT-4 and Llama-3 70B (KCC 2026)
- Sampling time-point introduced as a latent variable for temporal batch effects in tensile-strength prediction for biodegradable fiber spinning. With GBR, R^2 rose from 0.724 to 0.811, RMSE fell 17.2%, and accuracy within the industry tolerance rose from 87.1% to 92.4%. MOTIE/KIAT project P0022335 (UCWIT 2025)
- Methodology for building software-in-the-loop verification environments at lower cost (2026, ongoing)

2022.09–2024.06

Researcher **(reverse engineering)** Daegu, Korea

Arkdata

- Reverse-engineered transaction log formats across heterogeneous DBMSs (Oracle, PostgreSQL, MariaDB/MySQL, Tibero) to support development of a Change Data Capture (CDC) solution
- Designed and built a Kafka-based real-time pipeline across heterogeneous databases
- Implementation, analysis, and testing on a three-person research team

Teaching

2025.03–present

Teaching Assistant Daegu, Korea

School of Computer Science and Engineering, Kyungpook National University

- Introduction to Programming: Spring 2025, Summer 2025, Winter 2025–26, Summer 2026 (four terms)
- System Programming and Creative Convergence Design: Fall 2025
- Java Programming: Spring 2026

Projects

2025–present

ICT-ABB Convergence Major Project github.com/99JIK/ABB

Kyungpook National University

Real-time emotion classification from facial action units. Up to 30 faces at once, IoU-based multi-face tracking, event triggers on emotion changes and AU thresholds, FastAPI REST and WebSocket streaming. MediaPipe, TensorFlow.

2026.01–2026.03

ST Youngwon Smart Office github.com/99JIK/ST_YOUNGWON

Contract work, in production use

- RAG question answering over internal documents. Regulatory documents chunked hierarchically (article, paragraph, fixed size), retrieved by hybrid keyword and vector search
- LLM providers (OpenAI, Claude, Gemini, local Ollama) behind a protocol abstraction so they can be swapped, per the client's maintainability requirement
- FastAPI, ChromaDB, SQLite, Synology NAS (FileStation API) integration, JWT auth, Docker Compose deployment

2025.09–2025.12

FOCUS **(Facial Observation & Concentration Understanding System)**
github.com/99JIK/FOCUS

Team coursework project

Measures a learner's concentration from webcam video in real time. Scores distraction from five signals: eye closure, head orientation, gaze with head-rotation compensation, yawning, and facial action units. MediaPipe Face Landmarker, vanilla JS, Vite.

2025.06–present	**Provenance** **(code plagiarism detection)** *Personal project, Rust* Compares program structure rather than text, so renaming variables or rewriting comments does not hide a copy. Parses Python, Java, C, C++, JavaScript, TypeScript, and Go, with the parsers compiled into the binary so a grader installs nothing else. Starter code can be set as a template and discounted. Matches are reported as groups rather than pairwise, scored by similarity, containment, and density. Cross-platform desktop app. Built for the programming exams I assisted.	github.com/99JIK/Provenance

Technical Skills

Languages	Python, Java, C/C++, TypeScript, JavaScript
Web / Backend	FastAPI, Spring, Node.js, React, Vue.js
Data / ML	Oracle, PostgreSQL, MySQL, ChromaDB, scikit-learn, LightGBM
Tools	Docker, Git, Linux, Kafka